SHOULDER JOINT



INTODUCTION

The shoulder joint complex consists of 5 synergistic joints:

►**Three true joints**:

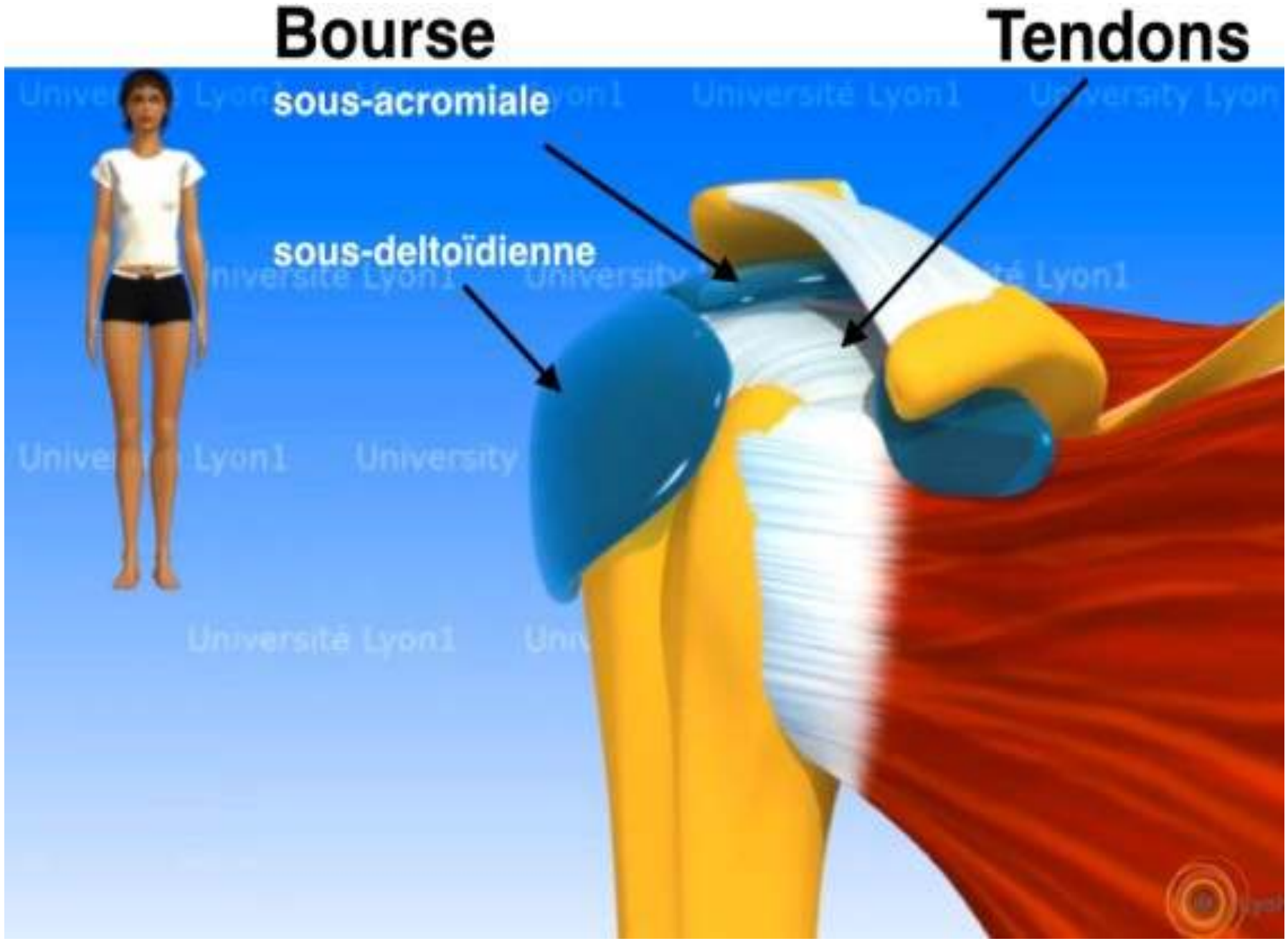
1. Scapulohumeral joint
2. Sternocostoclavicular joint
3. Acromioclavicular joint

► **Two sliding spaces**:

1- Serous burs

2- Scapulothoracic joint




Bourse
Tendons
sous-acromiale
sous-deltoïdienne

Vue ante,rieure,du complexe articulaire de l'epaule

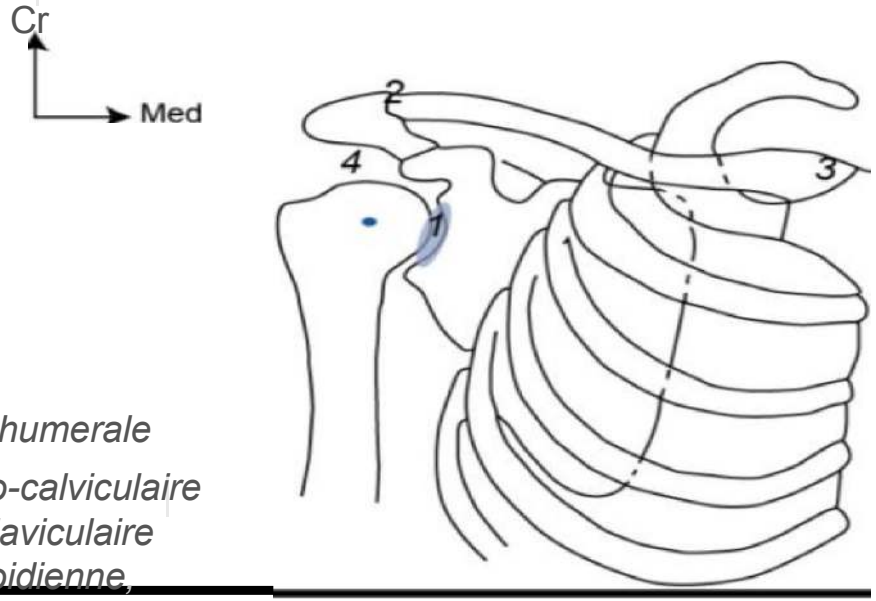


Figure 1

1. *Articulation scapulo-humerale*
2. *Articulation acrom[1]io-calviculaire*
3. *Articulation sterno-claviculaire*
4. *Articulation sub-deltoidienne,*

Articulation de repaule sur coupe transv:ersale

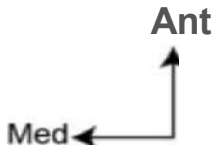
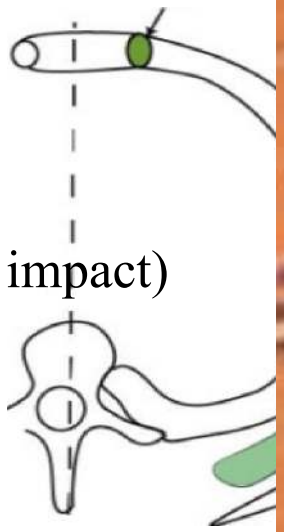


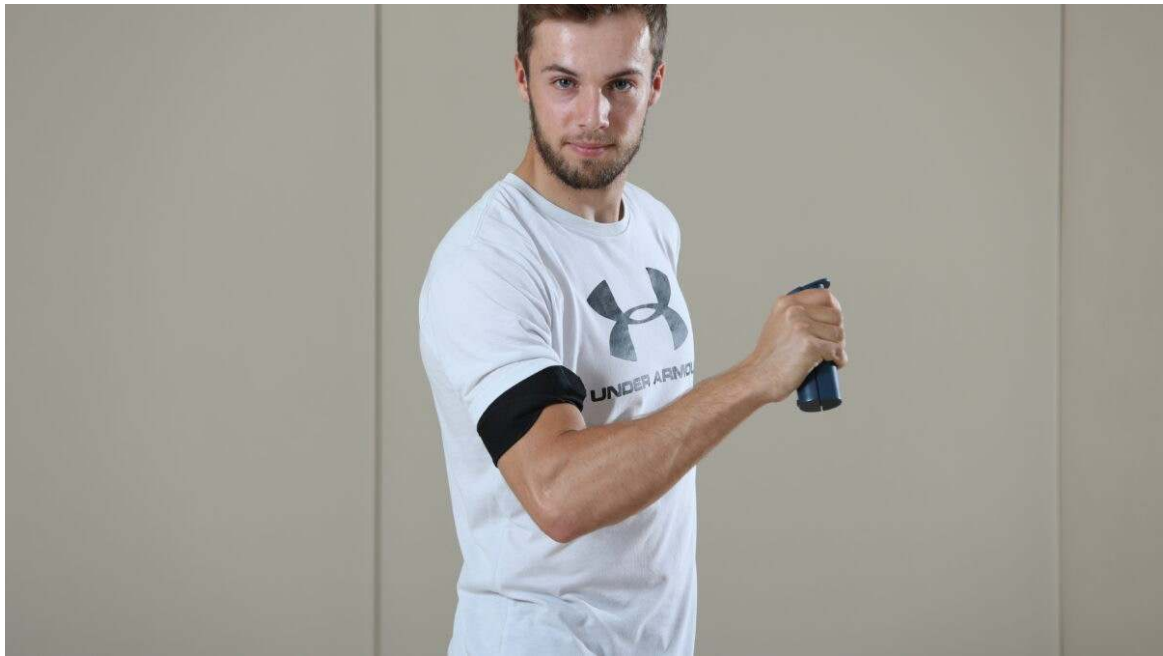
Figure 2

1.- *Articulation Sterno-alaviculaire*
2.- *Articulation A.crom to-claviculaire*
3.- *Articulation Scapulo -humer:ale*
4.- *Articlation Scapulo-thoracique*
5.- *Muscle deltoidien*

• The shoulder corresponds
to the proximal joint of the upper limb,
• It is a highly mobile joint,
• It is suspended, exposed, and superficial (prone to impact)



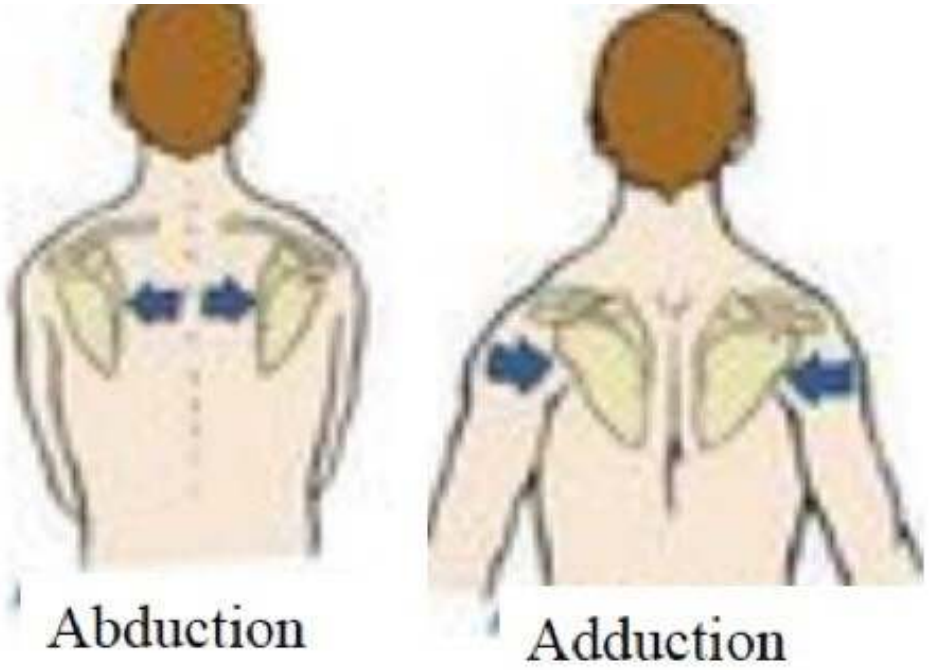


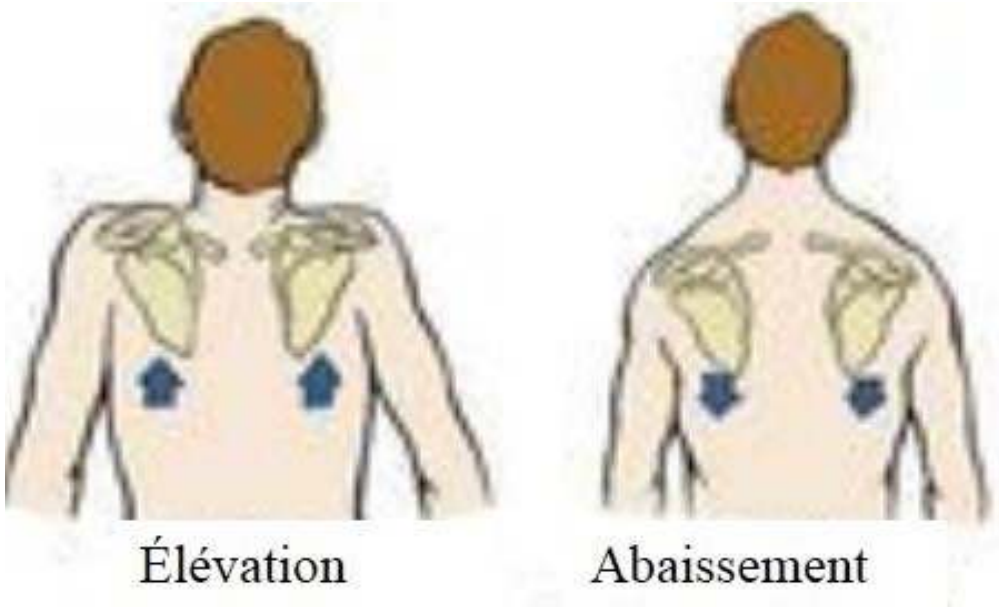


I- **Sternocostoclavicular joint**

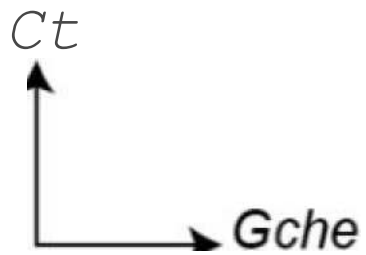
• It connects the medial end of the clavicle to the sternal manubrium and the first costal cartilage,
• It is a saddle synovial joint, with an intermediate articular disc.

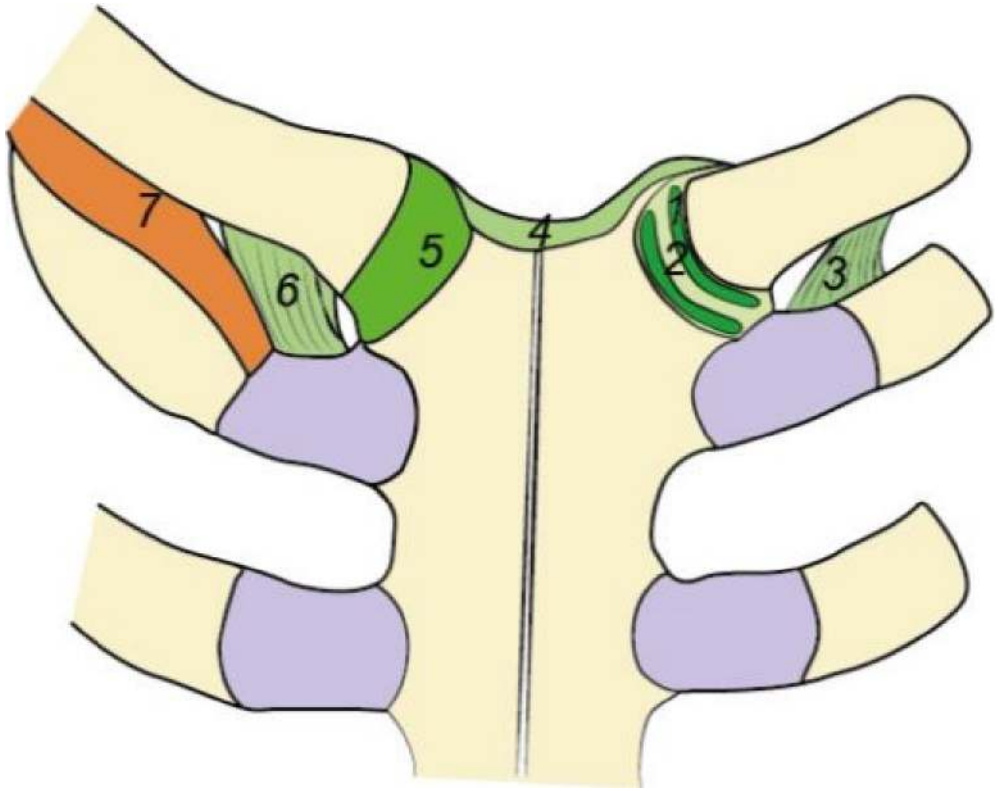
► Two types of movements of modest amplitude are possible:
Elevation, depression: results in elevation and depression of the shoulder.
Anterior projection, posterior projection.





Sternocostoclavicular joint





Vues antérieures

II- Acromioclavicular joint

• It is a flat surface joint: arthrodia type
• It connects the lateral end of the clavicle with the acromion.
• The surfaces are held in place by a capsule and the acromioclavicular ligament.

• It allows for very slight gliding movements.

III- Scapulothoracic joint:

• It is a false joint,
• it is a sliding space between the thoracic wall (the rib cage) and the anterior surface of the scapula

IV- Subdeltoid joint:

• Gliding of the deltoid over the scapulohumeral joint facilitated by the subacromiodeltoid serous bursa.

V- SCAPULOHUMERAL JOINT

1- DEFINITION

• Proximal joint of the thoracic limb; connects the humerus to the scapula

► Very mobile

► Extremely fragile

► Spheroid synovial joint (enarthrosis).

ARTICULAR SURFACES

I-1- The glenoid cavity of the scapula

• Oval in shape.

• It is surrounded on its peripheral edge by the glenoid labrum.

I-2- The humeral head: It occupies the proximal epiphysis of Rounded in shape, it represents 1/3 of a sphere with a diameter of 30 mm, the humerus,

- The glenoid labrum:

• It is a fibrocartilaginous ring attached to the rim of the glenoid cavity.

Means of union

1. The articular capsule
   - It is a thin fibrous sleeve
   - It inserts, on one hand, on the peripheral surface of the glenoid labrum and on the other hand, on the anatomical and surgical neck of the humerus.

2- Passive ligaments

2.1- The coracohumeral ligament: the strongest:

❖ It originates on the lateral border of the coracoid process.

❖ It terminates in two fascicles: one on the greater tubercle (trochiter), and the other on the lesser tubercle (trochin) of the humerus.

2.2- **The glenohumeral ligaments**

Superior: originates on the glenoid labrum and the superior part of the neck of the scapula and terminates on the superior part of the lesser tubercle (trochin) of the humerus.

Middle: originates anterior to the previous one and terminates on the inferior part of the lesser tubercle (trochin) of the humerus.

Inferior: originates on the glenoid labrum and the neck of the scapula and terminates below the surgical neck of the humerus

NB: These three ligaments delimit between them two spaces of capsular weakness:

The foramen ovale of Weitbrecht: between the superior and middle glenohumeral ligaments.
The foramen of Rouvière: between the middle and inferior glenohumeral ligaments

3- Active ligaments

• These are the tendons of the periarticular and mobilizer muscles of the scapulohumeral joint,

• These are the muscles:

• Supraspinatus: above

• Infraspinatus, teres minor and teres major: behind

• Subscapularis: in front.

Tendon of the long head of the biceps

The deltoid.

Tendon of the triceps muscle.

2. The synovial membrane

• It lines the deep surface of the articular capsule up to its insertions.

• The synovial fluid produced by the synovial membrane allows for the lubrication and nutrition of the articular cartilages.

3- Periarticular serous bursae

• There are a number of serous bursae between the capsule and the periarticular muscles whose role is to facilitate gliding; the most important are:

The subscapular serous bursa.
The bicipital serous bursa.
The subdeltoid serous bursa.
The subcoracoid serous bursa.
The serous bursa of the infraspinatus muscle

Functional anatomy

• The scapulohumeral joint is very mobile

• It possesses 3 degrees of freedom according to the 3 planes and the 3 axes.

The frontal plane and around the anteroposterior axis

• **Adduction**: It moves the limb inward, can reach 40°.

• **Abduction**: It moves the limb outward, amplitude 90°.

In the sagittal plane and around the transverse axis:

• **Flexion (antepulsion):** It projects the limb forward, amplitude 50-80°

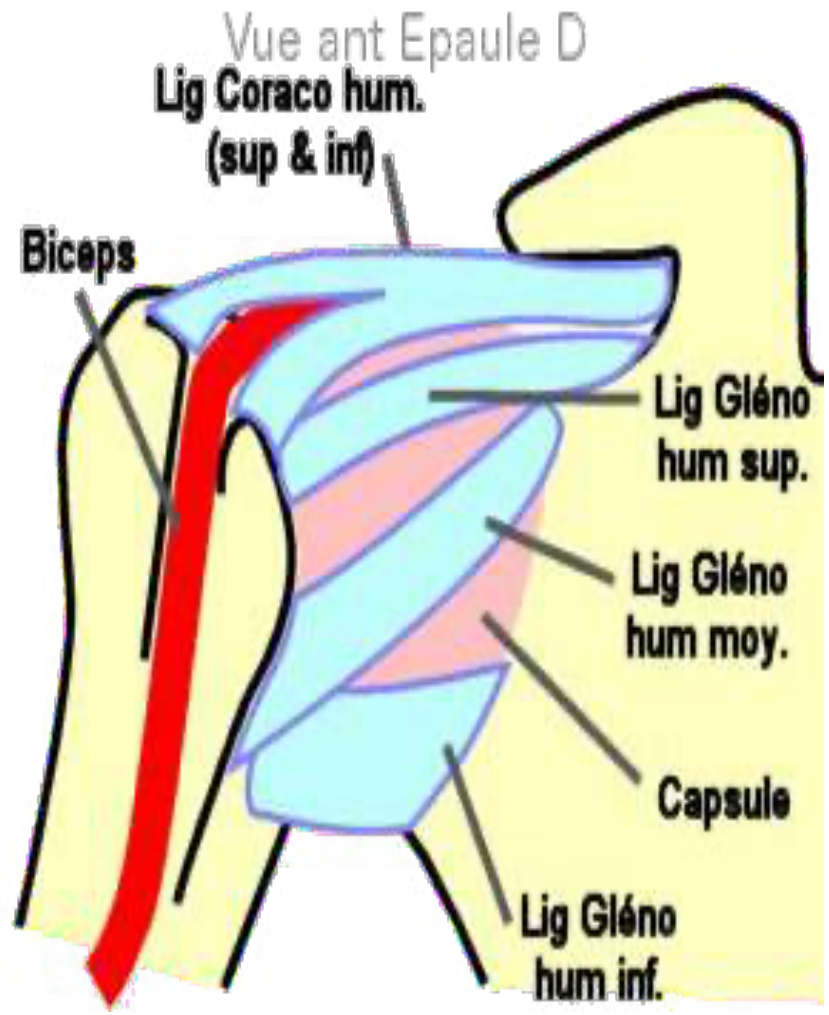
• **Extension (retropulsion):** It projects the limb backward, never exceeds 30°.

In the transverse plane and around a vertical axis

• **Lateral (external) rotation**: is limited to 30° amplitude.

• **Medial (internal) rotation**: is limited to 80°; an anterior tilt of the scapula completes it to 90°.

• **Circumduction**: It encompasses all types of movements and describes a cone with its apex passing through the center of the humeral head.


Vue ant Epaule D
Lig Coraco hum.
(sup & inf)
Biceps
Lig Gléno
hum sup.
Lig Gléno
hum moy.
Capsule
Lig Gléno
hum inf.

Mouvement	Amplitude (Épaule)	Illustration
Flexion / Extension (antépulsion / rétropulsion)	180° / 50°	 / 
Abduction / Adduction	180° / 50°	 / 
Rotation interne / externe	90° / 90°	 